MINI PROJECT REPORT

AI-Based Electricity Bill & Energy Consumption Analyzer

A Web Application Developed Using Base44 AI Tool

B.Tech – Electrical and Electronics Engineering (EEE)

Project Link:

https://watt-wise-ai.base44.app

GitHub-ready Mini Project Documentation

Academic Year: 2026

1. Abstract

The AI-Based Electricity Bill & Energy Consumption Analyzer is a web-based energy management application designed to help users understand, monitor, and reduce their electricity consumption. The project combines electrical engineering concepts with artificial intelligence and a no-code/AI application development platform, Base44. The application analyzes electricity usage, appliance-level consumption, estimated costs, energy trends, and provides AI-assisted recommendations for reducing energy wastage. It is intended as a practical demonstration project for B.Tech Electrical and Electronics Engineering students.

2. Project Information

3. Introduction

Electricity consumption is an important concern for households, educational institutions, offices, and small businesses. Users often receive electricity bills without having a clear understanding of which appliances are responsible for high consumption or how changes in usage affect the bill. This project addresses that problem by providing a simple digital platform for electricity bill analysis and energy consumption monitoring.

The application uses electrical energy calculations together with AI-assisted analysis to convert raw usage information into understandable insights. Users can enter electricity bill data, record appliance ratings and usage hours, view consumption trends, estimate costs, and receive energy-saving suggestions.

4. Problem Statement

Many electricity consumers do not have an easy-to-use system for analyzing their electricity bills, identifying high-energy-consuming appliances, comparing historical consumption, and receiving personalized energy-saving recommendations. Manual calculations can be time-consuming and may not clearly reveal energy wastage. Therefore, there is a need for an accessible web application that combines electrical energy calculations, visualization, and AI-based recommendations.

5. Objectives

- Develop a user-friendly web application for electricity bill and energy consumption analysis.

- Calculate appliance-wise and overall electricity consumption in kWh.
- Estimate electricity costs using user-provided consumption and tariff information.
- Visualize daily, monthly, and appliance-level energy usage.
- Identify unusually high consumption and possible energy wastage.
- Provide AI-assisted personalized energy-saving recommendations.
- Support comparison of current and previous electricity consumption.
- Demonstrate the use of an AI application development tool (Base44) in an EEE project.

6. Proposed Solution

The proposed solution is a responsive web application named VoltAnalyzer. The user enters electricity bill information and appliance details. The application processes the data using electrical engineering formulas, presents the results through dashboards and charts, and uses AI-assisted logic to generate recommendations and insights.

7. Key Features

8. Technology Stack

- Base44 AI – AI-assisted web application development platform.
- Web UI – Responsive browser-based interface.
- AI/Intelligent Analysis – Used for generating energy insights and recommendations.
- Data Storage – Application data structures supported by the Base44 environment.
- Charts and Visualizations – Used for energy trends, comparisons, and appliance analysis.
- GitHub – Recommended platform for source/documentation sharing and project presentation.

9. System Workflow

1. User opens the VoltAnalyzer web application.
2. User enters electricity bill details and/or appliance information.
3. The application calculates energy consumption and estimated cost.
4. Historical information is used for comparison and trend analysis.
5. The AI analysis interprets the available consumption information.
6. The application displays recommendations, warnings, and potential saving opportunities.
7. The user can review trends and use the results to make energy-efficient decisions.

10. Important Electrical Calculations

The application is based on standard energy-consumption relationships. The following formulas are used conceptually:

- Energy Consumption (kWh) = Power (W) × Usage Hours × Number of Days ÷ 1000
- Estimated Cost = Energy Consumption × Applicable Tariff
- Average Daily Consumption = Monthly Units ÷ Number of Billing Days
- Percentage Change = ((Current Consumption – Previous Consumption) ÷ Previous Consumption) × 100

Actual electricity tariffs vary by location, consumer category, billing provider, slab, taxes, and other charges. Therefore, cost calculations should be treated as estimates unless the exact tariff structure is supplied.

11. AI Component

The AI component is designed to turn consumption information into useful explanations and recommendations. For example, if an appliance contributes a large share of monthly consumption, the application can highlight it and suggest reducing operating hours or improving usage practices. AI-generated predictions and recommendations should be treated as decision-support information rather than guaranteed forecasts.

12. Base44 AI Development Approach

Base44 AI was used to accelerate development of the web application. The project requirements were expressed as natural-language instructions covering the application purpose, pages, calculations, dashboard, forms, visualizations, and AI features. The resulting application was then organized as a practical energy-management prototype.

8. Define the project problem and target users.

9. Provide Base44 with a detailed application-generation prompt.

10. Generate the application structure and user interface.

11. Add electricity, appliance, and energy-analysis functionality.

12. Test calculations and user flows.

13. Refine the interface and content for an EEE academic demonstration.

14. Publish the application through its Base44 deployment link.

13. User Interface / Screens

For the GitHub repository, add screenshots of the actual application below. Recommended screenshots include:

- Home/Landing Page
- Dashboard
- Bill Analyzer
- Appliance Analyzer
- Energy Trends
- AI Energy Advisor
- Bill Prediction
- Energy Saving Simulator
- Reports / Alerts

Suggested GitHub screenshot naming:

- screenshots/home.png
- screenshots/dashboard.png
- screenshots/bill-analyzer.png
- screenshots/appliance-analyzer.png

- screenshots/ai-advisor.png

14. Expected Results

- Users can understand their electricity consumption more clearly.
- Appliance-level energy usage can be estimated using power and usage information.
- Users can identify high-consumption appliances.
- Historical consumption can be compared through charts.
- AI-assisted recommendations can help users identify possible energy-saving actions.
- The project demonstrates how AI tools can be applied to a practical EEE problem.

15. Advantages

- Simple and user-friendly web interface.
- Combines EEE knowledge with AI.
- Reduces manual calculation effort.
- Provides visual energy-consumption insights.
- Useful for household and small-business energy awareness.
- Can be extended with IoT and smart-meter data in future.

16. Limitations

- Accuracy depends on the quality of user-entered data.
- Bill estimates may differ from actual bills because tariff structures vary.
- AI predictions are estimates and should not be treated as guaranteed values.
- Real-time appliance monitoring requires suitable sensors or smart-meter integration, which is outside the basic web prototype.

17. Future Scope

- Integrate IoT energy meters for real-time voltage, current, power, and energy monitoring.
- Connect with smart plugs and smart-home devices.
- Add weather data for improved solar/energy analysis.
- Develop machine-learning models using larger historical datasets.
- Add mobile application support.
- Add regional electricity tariff databases.
- Provide automated alerts through email or mobile notifications.
- Integrate solar PV generation and battery-storage analysis.

18. Conclusion

The AI-Based Electricity Bill & Energy Consumption Analyzer demonstrates a practical application of Electrical and Electronics Engineering concepts with modern AI-assisted web development. The project helps users analyze electricity consumption, understand appliance-level usage, estimate costs, observe trends, and receive energy-saving guidance. Using Base44 AI makes it possible to rapidly develop and present a functional prototype while focusing on the underlying EEE problem and user value.

19. Project Link

Live Application: https://watt-wise-ai.base44.app

20. GitHub Repository Structure

- README.md
- Mini_Project_Report.docx
- screenshots/
- screenshots/home.png
- screenshots/dashboard.png
- screenshots/bill-analyzer.png
- screenshots/appliance-analyzer.png
- screenshots/ai-advisor.png
- LICENSE (optional)

21. Suggested GitHub README Description

VoltAnalyzer is an AI-based web application developed as a B.Tech EEE mini project using Base44 AI. It analyzes electricity bills and energy consumption, provides appliance-wise usage insights, visualizes energy trends, estimates costs, and offers AI-assisted recommendations for reducing energy consumption.

Project Title	AI-Based Electricity Bill & Energy Consumption Analyzer
Application Name	VoltAnalyzer
Project Type	B.Tech EEE Mini Project
Development Platform	Base44 AI
Application Type	Responsive Web Application
Domain	Electrical Energy Management / AI
Currency	Indian Rupee (■)
Energy Unit	kWh / Units
Live Project	https://watt-wise-ai.base44.app

Feature	Description
Dashboard	Displays key energy and cost indicators in a single view.
Bill Analyzer	Analyzes units consumed, bill amount, cost per unit, and changes in usage.
Appliance Analyzer	Calculates appliance-wise energy consumption and estimated cost.
Energy Trends	Presents historical consumption using charts and comparisons.
AI Energy Advisor	Provides personalized recommendations based on usage patterns.

Bill Prediction	Provides an estimated future bill when sufficient historical information is available.
Energy Saving Simulator	Compares current and proposed appliance usage to estimate savings.
Carbon Footprint	Provides an estimated electricity-related carbon-emission view.
Alerts	Highlights unusually high consumption or budget-related conditions.
Reports	Organizes important monthly energy information for review.